

THAT'S NOT ENOUGH

FORECASTING DENGUE BEYOND THE LEADERBOARD

Why a better score is not enough

From prediction accuracy to decisions that affect patients

WE HAVE TO GO DEEPER

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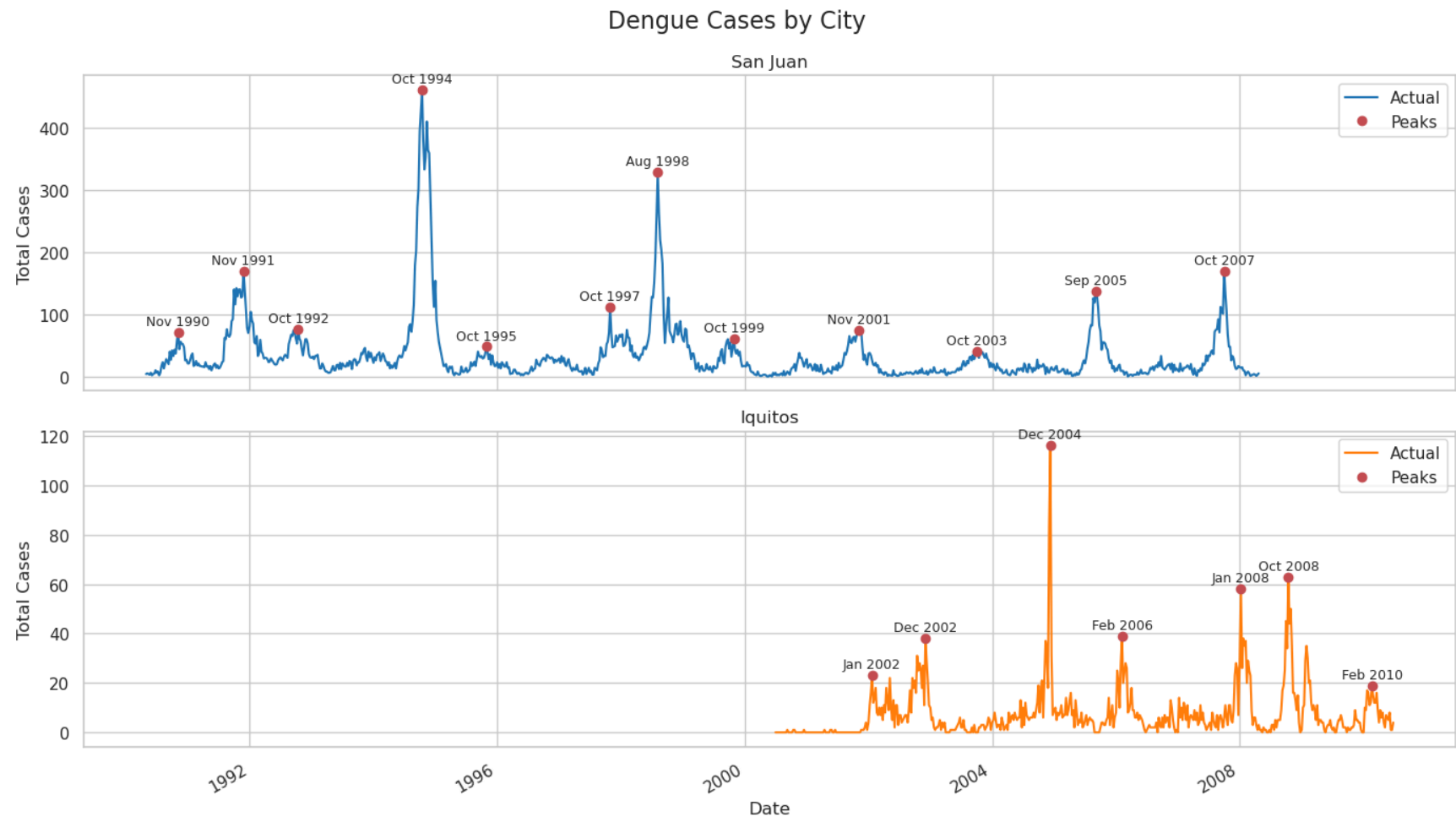
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	HOLDOUT MAE
sarimax	4.61
catboost	6.27

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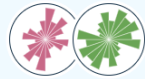
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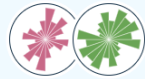
1w 2d ago · 14 submissions

23.6010

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Because we did not want to stretch the owl over the globe! At some point, something started to feel wrong.



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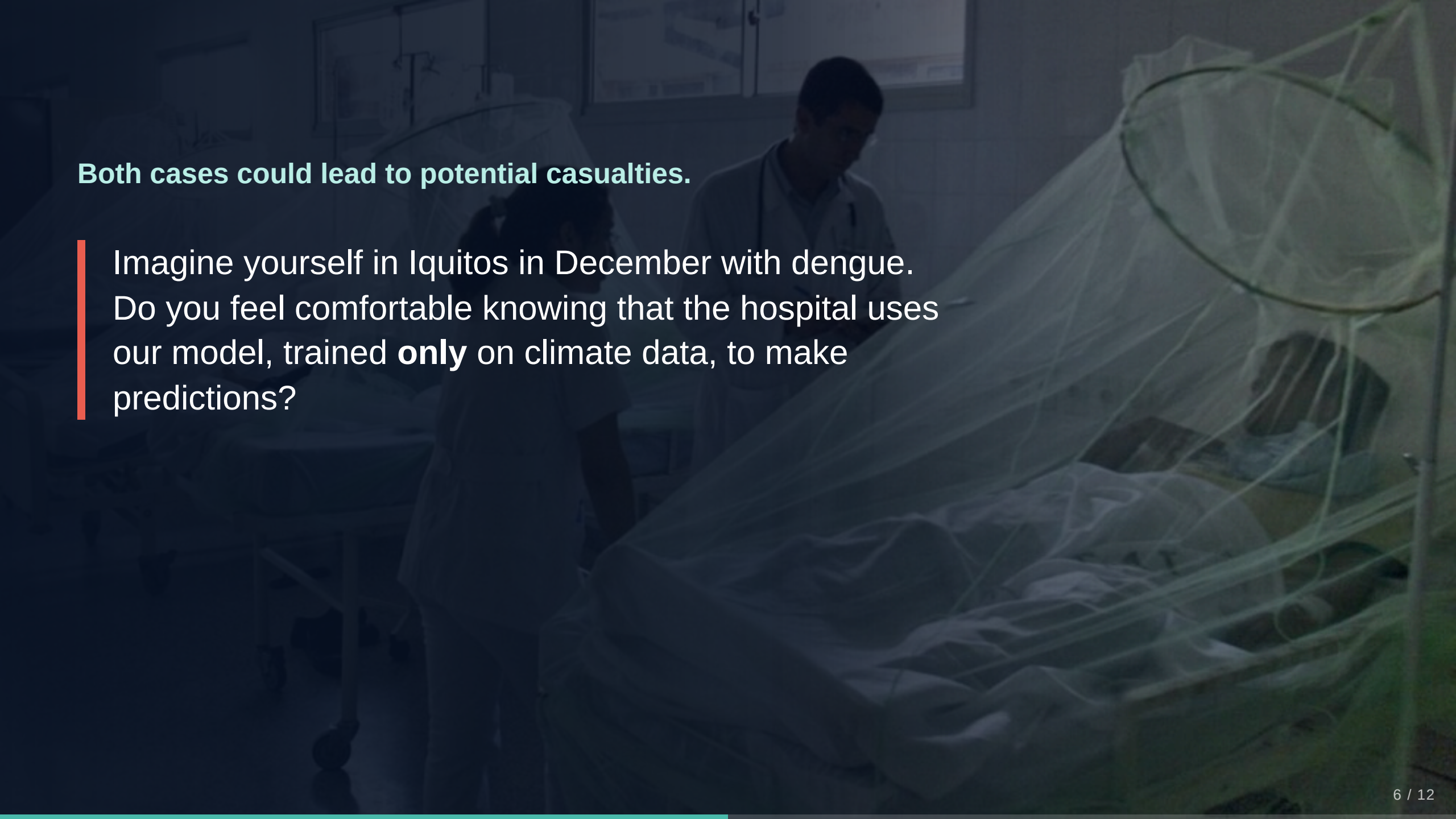
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Underprediction: The hospital prepares fewer medications than needed, directly risking patient safety and care quality.

Overprediction: The hospital over-allocates resources, leading to expired supplies, financial waste, and strained capacity in other departments.



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Imagine yourself in Iquitos in December with dengue. Do you feel comfortable knowing that the hospital uses our model, trained **only** on climate data, to make predictions?

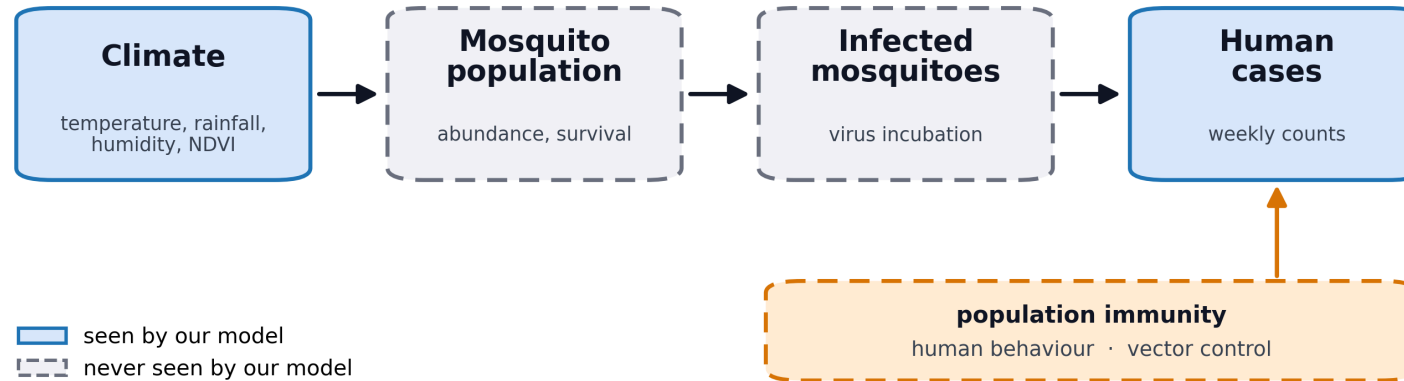
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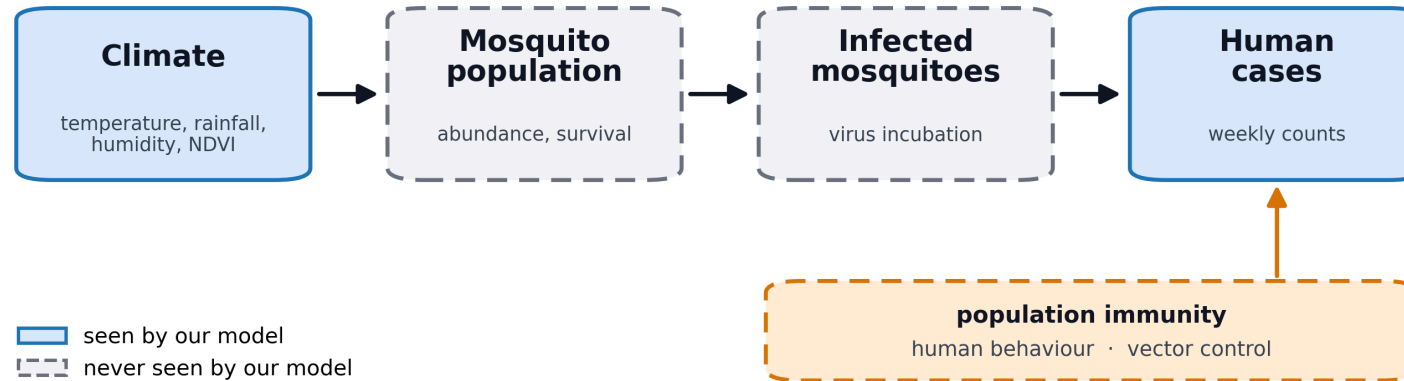
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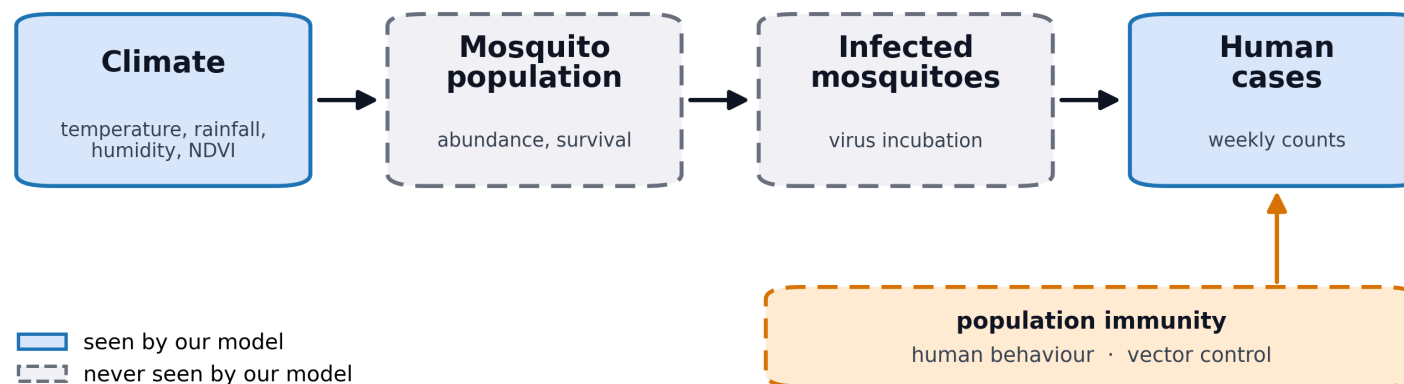
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Predicting a curve is not the same as understanding what moves it. A model that has never seen a mosquito cannot tell us what happens when the mosquitoes change.

Source: “Long-Term and Seasonal Dynamics of Dengue in Iquitos, Peru”, PLoS NTD, DOI 10.1371/journal.pntd.0003003 (10 years of laboratory-confirmed surveillance, 2000-2010).

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Transmission was dominated by single serotypes, first DENV-3 (2001-2007) then DENV-4 (2008-2010). After 2003, incidence fluctuated inter-annually with outbreaks usually occurring between October and April. We detected a strong positive autocorrelation in case counts at a lag of ~70 weeks, indicating a shift in the timing of peak incidence year-to-year. All climatic variables showed modest seasonality and correlated weakly with the number of reported dengue cases across a range of time lags. Cases were reduced after citywide insecticide fumigation if conducted early in the transmission season.

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Conclusions

Dengue case counts peaked seasonally despite limited intra-annual variation in climate conditions. Contrary to expectations for this mosquito-borne disease, no climatic variable considered exhibited a strong relationship with transmission. Vector control operations did, however, appear to have a significant impact on transmission some years. Our results indicate that a complicated interplay of factors underlie DENV transmission in contexts such as Iquitos.

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Some of the data is accesible only by request and we had no time to email the authors.

The presence of such continued studies serve as an indicator that the experts in the field actually understand the necessity for the stronger correlated data.

Insights from [Barrera et al., 2019](#) say that

Severe storms are expected to increase the number of mosquitoes by increasing or decreasing the edge surface of groundwater habitats, filling cavities/containers with rainwater, and generating additional container aquatic habitats in destroyed or abandoned properties (damaged appliances, inundated floors/roofs, and rubble).

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We could not find any studies exploring the causal link between the oil spill and the dengue outbreak that year.

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The model is likely to forecast the curve. We cannot yet say that it understands why the curve moves, and in a hospital, this is what matters the most.

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Questions?